

TICKET BOOKING ANALYSIS

1. Project Overview:

The project is based on the problem statement of analyzing entertainment booking data to understand booking trends, audience preferences, revenue patterns, and high-demand periods. The data is cleaned, transformed and analyzed using Excel, while Power BI is used to present the findings through an interactive dashboard.

Project Objective:

The main objective is to use the entertainment booking data to identify patterns and trends related to customer preferences, ticket booking, movie performance and revenue. The analysis aim to generate meaningful insights that address the problem statement and support better business decision-making.

2. Data Source:

Source: ChatGPT – generated synthetic dataset.

Description: The dataset contains movie, cinema and entertainment booking information including movie details, cinema location, booking dates, show dates, ticket prices, payment methods, customer information and revenue.

Timeline: 2026(Booking and Show Dates)

Movie Release Years: 2015 – 2024.

Domain: Entertainment/Movie Ticket Booking/Business Analysis.

3. Problem Statement:

To analyze entertainment booking data across movies, cities, cinemas, customers groups and payment methods to identify booking trends, audience preferences, movie performance revenue patterns and high-demand period for better decision-making.

4. Attributes(Column/Features)Details:

Attribute Name	Data Type	Description
Booking ID	String	Unique identifier for booking.
Movie ID	String	Unique identifier for movie.
Movie Name	String	Name of the movie booked.
Genre	Categorical	Genre of the movie such as Action, Comedy, Adventure or Thriller.
Language	Categorical	Language of the movie.
Rating	Numeric	Rating given to the movie.
Cinema Name	String	Name of the theatre where the movie is screened.
City	Categorical	City where the cinema is located.
State	Categorical	State where the cinema is located.
Booking Date	Date	Date on which the ticket was booked.
Show Date	Date	Date on which the movie show takes place.
Show Time	Time	Time of the movie show.
Ticket Price	Numeric	Price of the ticket for the booking.

Total amount	Numeric	Total amount generated from the booking.
Payment Method	Categorical	Payment method used for the booking such as UPI, Credit Card, Debit Card, Net Banking or Wallet.
Age	Numeric	Age of the customer making the booking.
Age Group	Categorical	Age category such as Younger, Middle Age and Above 45.
Booking Category	Categorical	Booking category classified as High, Medium and Low.
Day Type	Categorical	Indicates whether the booking/show date is a Weekday or Weekend.
Month	Categorical	Month in which the booking or show takes place, used for trend analysis.

5. Tools and Technologies:

Excel:

- Data cleaning and preprocessing.
- Removing duplicate records.
- Checking and handling missing values.
- Data transformation and formatting.
- Creating calculated columns such as Age Group, Day Type and Month.

- Using lookup functions to combine relevant data.
- Pivot tables for booking, revenue, city, age-group, payment and movie analysis.

Power BI:

- Data import and transformation.
- Data modeling and relationship creation.
- DAX calculations and measures.
- KPI cards and interactive visualizations.
- Bar charts, column charts, Donut charts, tables and maps.
- Slicers and filters for interactive analysis.
- Interactive dashboard creation.

Identifying trends and generating business insights.

6. Data Preprocessing: (Excel & Power BI)

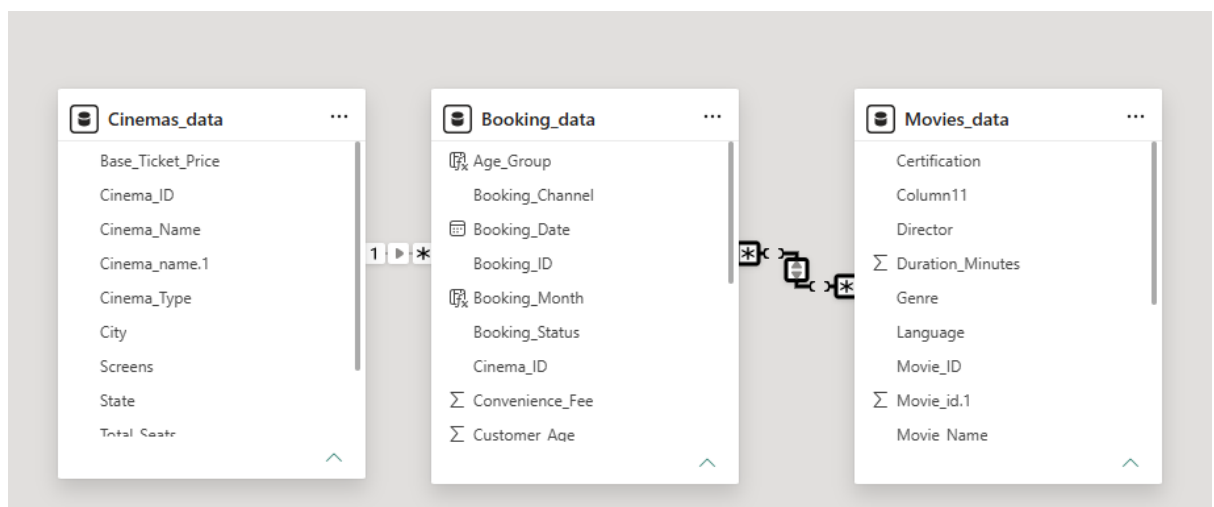
- **Data Cleaning & Transformation:** Removed duplicate records, checked for blank values, standardized date and time formats and created calculated fields such as Age Group, Day Type, Month and Revenue.
- **Filtering & Sorting:** Filtered and sorted the data to identify and organize relevant booking records.
- **Data Validation:** Checked Booking IDs, duplicate records, blank values and inconsistent data before analysis.
- **Data Integration:** Used Movie Data, Cinema Data and Booking Data together for analysis using common fields such as Movie ID and Cinema ID.
- **Pivot Tables:** Created Excel Pivot Tables to summarize bookings, revenue, ticket price, city, theater, payment method, age group and day type.
- **Power BI Data Preparation:** Imported the cleaned Excel tables into Power BI Desktop and prepared the data for relationships calculations and visualization.

7. Data Modelling (DAX & Power BI):

- **Data Model:** Established relationships between the Booking, Movie and Cinema tables using common fields such as Movie ID and Cinema ID.
- **Relationships:** Defined appropriate relationships between tables to enable accurate analysis across different datasets.
- **Data Structure:** Organized the tables in Power BI Model view to support efficient analysis and visualization.
- **DAX Calculations:** Created measures and calculated fields for key metric such as Total Revenue, Booking Count, Average Ticket Price and Average Rating.
- **KPI Measures:** Created measures to display important KPIs in dashboard cards and other visuals.

Screenshot of Data Modelling:

- **Power BI Model View:**



- **Calculated Columns:**

Payment_Mode	Booking_Status	Customer_Age	Customer_Gender	Booking_Channel	Format	Booking_Month	Age_Group	Revenue	Day Type
credit Card	Confirmed	30	Male	Mobile App	2D	April	Middle Age	480	Weekday
credit Banking	Confirmed	31	Female	Website	2D	March	Middle Age	480	Weekday
credit Banking	Confirmed	23	Other	Mobile App	2D	February	Youngster	480	Weekday
credit Banking	Confirmed	21	Male	Mobile App	2D	May	Youngster	480	Weekday
credit Banking	Confirmed	53	Female	Mobile App	2D	February	Above 45	480	Weekday
credit Banking	Confirmed	27	Female	Mobile App	2D	March	Youngster	480	weekend
credit Card	Confirmed	60	Other	Mobile App	2D	April	Above 45	480	weekend
credit Banking	Confirmed	30	Other	Mobile App	2D	June	Middle Age	480	Weekday
credit Banking	Confirmed	44	Female	Website	2D	February	Middle Age	480	Weekday
credit Card	Confirmed	55	Other	Mobile App	2D	April	Above 45	480	Weekday

Calculated columns such as Age Group, Booking Month, Revenue, and Day type were created to support detailed analysis of customer demographics, booking trends, revenue and weekday/weekend booking patterns.

Age Group:

```
1 Age_Group = IF(Booking_data[Customer_Age]<18,"Below 18",if(Booking_data[Customer_Age]<30,"Youngster",IF(Booking_data[Customer_Age]<45,"Middle Age","Above 45")))
```

Revenue:

```
1 Revenue = Booking_data[Tickets]*Booking_data[Ticket_Price]
```

Day Type:

```
1 Day Type = if(weekday('Booking_data'[Booking_Date],2)<=5,"Weekday","weekend")
```

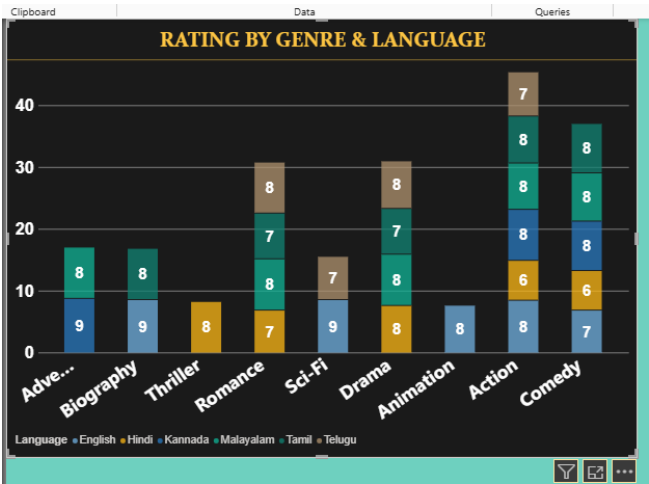
8. Analysis and Visualization:

- KPI Cards:



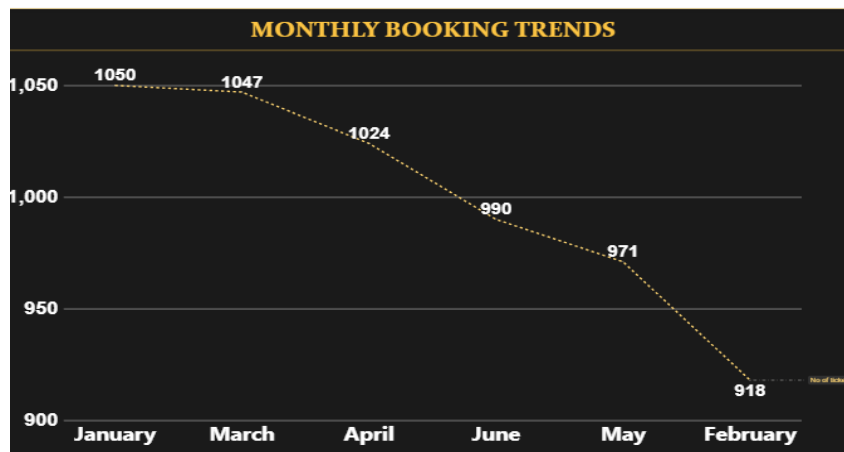
KPI Cards showing key entertainment booking metrics.

- Rating by Genre & Language:



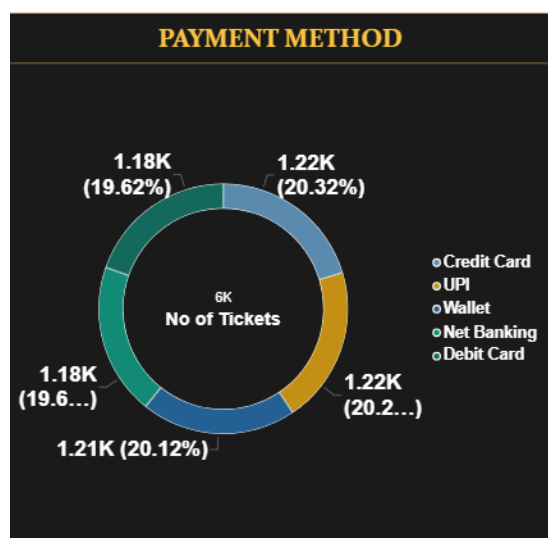
The chart shows the average movie rating across different genres and languages. Action movie recorded the highest average rating among the genres analyzed. The comparison also shows differences in rating across languages, helping identify genres and languages associated with higher audience ratings.

- **Monthly Booking Trends:**



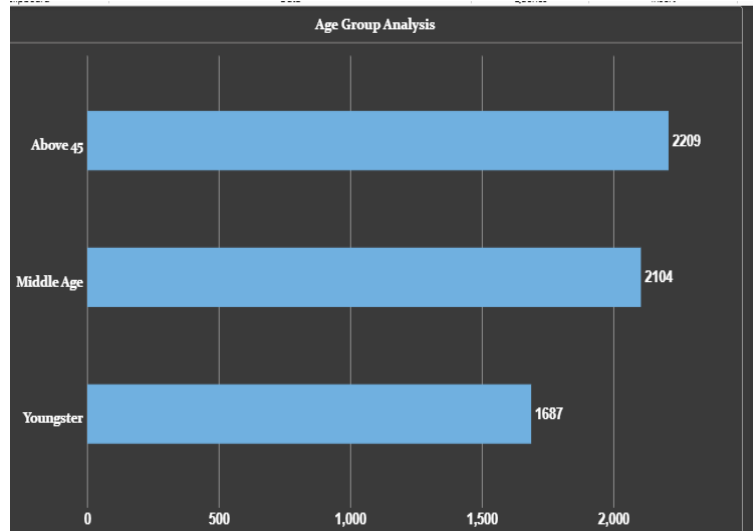
Monthly booking trends show variations in booking demand across different months, helping identify periods of higher and lower booking activity.

- **Payment Method:**



Credit card recorded the highest number of bookings, while Debit Card, Net Banking, UPI and Wallet showed relatively similar booking volumes.

- **Age Group Analysis:**



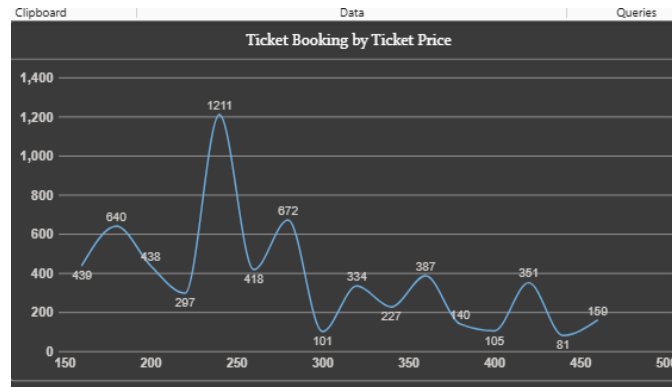
The Above 45 age group recorded the highest number of bookings, followed by the Middle Age group, while the Younger group had the lowest bookings.

- **Map Analysis:**



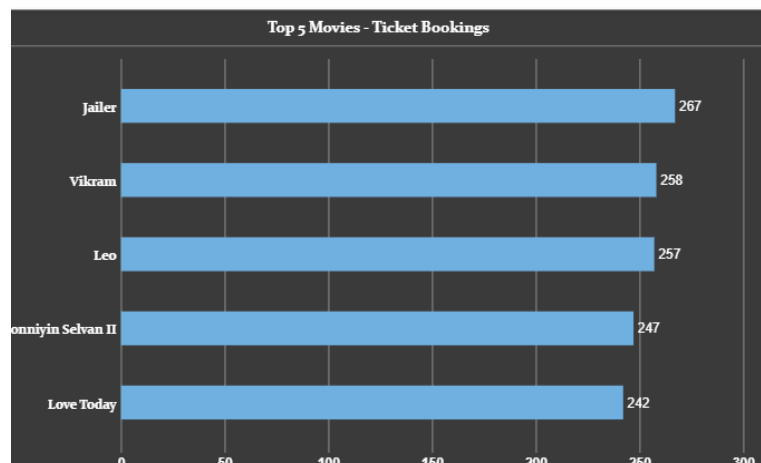
Pune recorded the highest bookings, while Kolkata and Kochi also showed strong booking activity.

- **Ticket Booking by Ticket Price:**



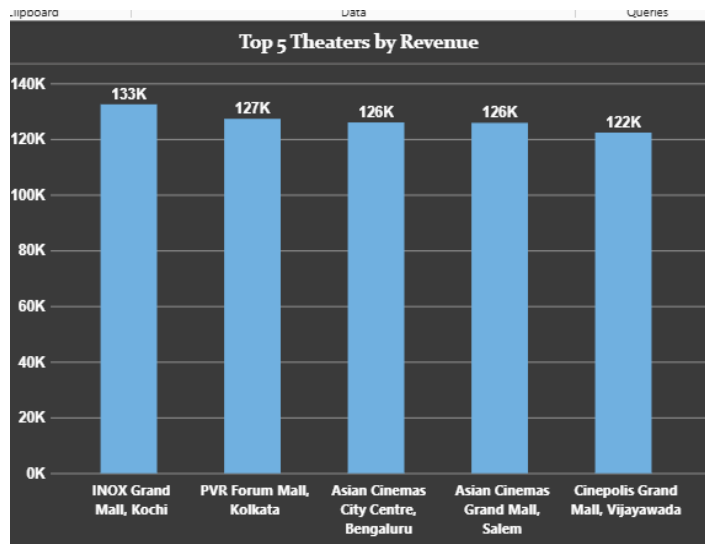
Ticket bookings varied across ticket price ranges, showing differences in customer demand at different price levels.

- **Top 5 Movies - Ticket Booking:**



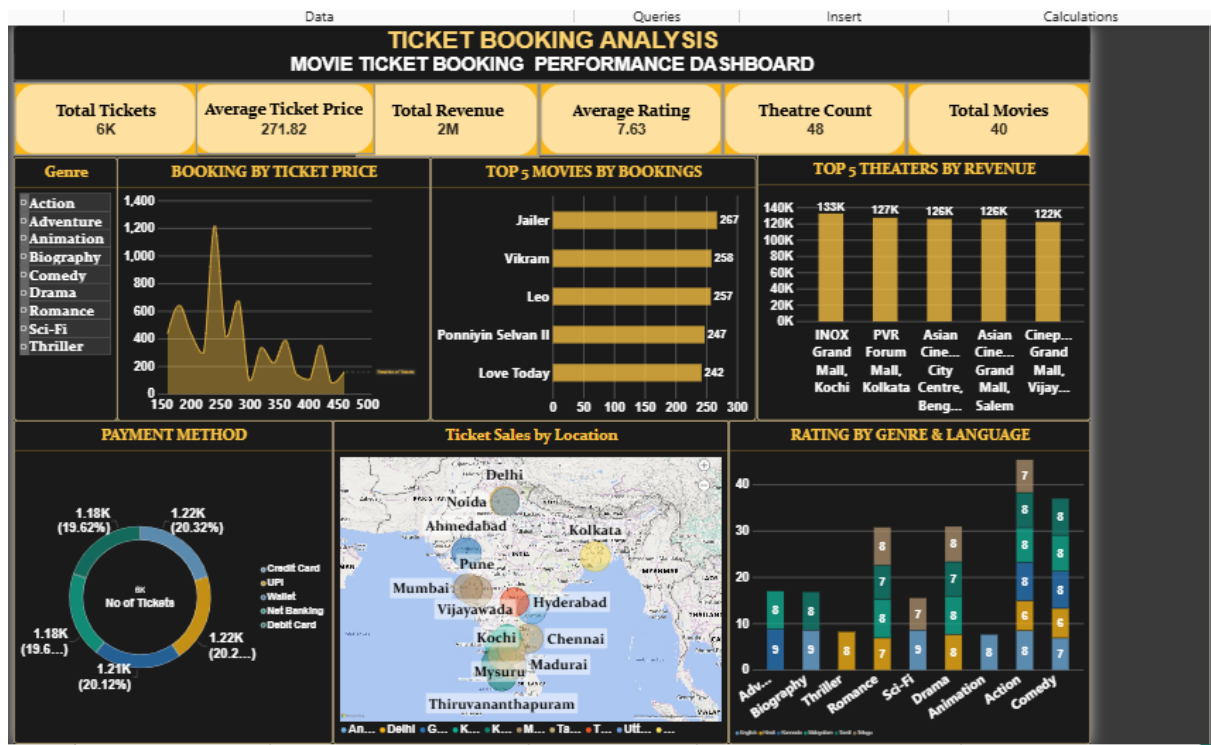
The top five movies recorded the highest number of ticket bookings, indicating strong audience interest and demand for these movies.

- **Top 5 Theaters by Revenue:**



The top five theaters generated the highest revenue, with PVR Forum Mall recording the highest revenue among them.

- **Consolidate Dashboard:**



The consolidated dashboard provides an overall view of booking trends, revenue, customer preferences, movie performance, payment methods and geographic patterns, helping identify key entertainment booking insights and high-demand areas.

9. Insights & Conclusion:

- **Descriptive Analysis:** The analysis summarizes booking volume, revenue, ticket prices, customer age groups, payment methods, cities, theaters, genres and movie performance. Power BI visualizations were used to identify major trends and patterns in the entertainment booking data.
- **Diagnostic Analysis:** The data was compared to understand factors associated with booking activity and revenue. The analysis showed differences in bookings based on age group, city, payment method, day type, movie and theater performance.
- **Predictive Analysis:** Predictive analysis was not performed in this project because no forecasting or machine learning model was developed. The dashboard focuses mainly on historical booking data and existing trends.
- **Prescriptive Analysis:** Based on the identified trends, theaters and entertainment businesses can focus on high-demand cities, popular movies and preferred customer segments. Promotional offers and targeted campaigns can also be planned during lower-demand periods to improve bookings and revenue.

10. Conclusion:

The integration of Excel and Power BI proved effective for analyzing entertainment booking data from data cleaning and transformation to interactive visualization. The analysis helped identify booking trends, customer preference, theater revenue, payment methods and geographic booking patterns. The Power BI dashboard provides a clear and interactive view of the findings, helping entertainment businesses understand customer demand and

make better decisions related to promotions, pricing and move performance.

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